

Data Architecture Map Annotation Report

One Earth Balance Sheet — Malaysia 2026

Executive Summary

This document explains the updates and the five new columns added to `Data_Architecture_Map_yk_ANNOTATED.xlsx`, which annotate all 788 application questions with their purpose, collection frequency, priority level, potential data sources, and quality constraints. Additionally, it highlights newly mapped balance sheet liabilities.

What Was Annotated & Updated

- All 788 questions in the data architecture map have been systematically annotated with five key attributes:
- **Relevance:** What each question is used for (wellbeing scoring, balance sheets, administration, context)
- **Frequency_of_collection:** How often this data should be collected
- **Priority:** Importance level for project goals (P1-P4)
- **Data_Source:** Potential secondary or external data sources to validate or complement collected data
- **Quality_Constraint:** Known limitations or biases that might affect data accuracy
- Additionally, the "Household" sheet has been updated so that "Aspirations", "Wishes", and "Unfulfilled Needs" (Q0243-Q0265) explicitly link to the Village Balance Sheet as "BS - Liability".

Understanding the New Columns & Mappings

1. Relevance — What Is This Question Used For?

This column shows the primary purpose of each question in the project.

Code	Category	Description	Example
Wellbeing	Wellbeing Scoring	Direct input to individual wellbeing score (7 spheres)	Height/Weight → BMI (Physical)
Balance Sheet	Balance Sheet	Village-level resource tracking (11 domains)	Land inventory → Land Balance Sheet
Wellbeing + Balance Sheet	Dual Purpose	Serves both wellbeing AND balance sheet	Water quality → Environmental + Water Balance Sheet
Wellbeing(derive)	Derivable	Can compute wellbeing indicators from this data	Food consumption → Dietary diversity
CONTEXT	Contextual	Qualitative context for planning	Aspirations, wishes, community profile

Code	Category	Description	Example
ADMIN	Administrative	Registration and setup only	Village name, household ID, moderator

2. Frequency_of_collection — How Often to Collect?

This column indicates the recommended data collection interval.

Frequency	Count	When to Use
Annually	743 questions (94.3%)	All operational data: production, consumption, resources, wellbeing
Once	24 questions (3.0%)	Registration and demographic baseline (one-time setup)
Every 2-3 years	21 questions (2.7%)	Aspirational data that changes slowly (wishes, long-term goals)

3. Priority — How Important Is This Question?

This column ranks questions by importance to project goals.

Level	Name	Description	Examples
P1	Critical	Essential for wellbeing scoring or balance sheet calculation	Income, food consumption, water quality, education
P2	Important	Significantly enhances scoring accuracy	Storage methods, business profitability, housing amenities
P3	Contextual	Valuable for intervention planning and gap analysis	Skills inventory, aspirations, land requirements
P4	Setup / Low Priority	One-time setup or secondary information	Registration, community profile (languages, hobbies)

4. Data_Source — Where Else Can We Get This?

Lists alternative or supplementary databases where this information might already exist.

Potential Source	Used For
National ID / Census Database	Demographic and Household details
Local Water Authority / Testing	Water quality and sanitation parameters
Agricultural Dept / Remote Sensing	Land, farming, and forestry data
National Grid / Utility Bills	Electricity and grid usage data

5. Quality_Constraint — Proceed With Caution

Outlines biases or practical hurdles in collecting accurate data.

Constraint Type	Example Application
Subjective Reporting / Recall Bias	People might misremember exact food amounts consumed
PII / Privacy Concerns	Income data, debt burden, identity document numbers
Observation Required	Housing area footprint, true property boundaries
Equipment Dependent	Blood pressure, detailed water chemical testing

6. Structural Updates: Liabilities Mapping

The primary data sheet was formally renamed to "Household". Furthermore, previously subjective questions covering "Aspirations", "Wishes", and "Unfulfilled Needs" (Q0243-Q0265) were mapped as direct Balance Sheet Liabilities ("Balance Sheet - Liability" relevance and "P2 - Liability for balance sheet" priority). This ensures community gaps are treated as mathematical deficits needing resolution rather than mere narrative context.

Annotation Statistics

Relevance Breakdown (Top 15 Categories)

Category	Count
BS - Education / Infrastructure	180
BS + WB(derive)	154
BS - Community Infrastructure	63
BS - Food	60
BS	33
BS - Water	30
BS - Liability	23
BS - Energy	23
BS - Food/Land	21
BS - Food/Income	20
ADMIN	19
CONTEXT	18
BS - Forestry	15
WB - Cultural / Social	13
BS - Mobility / Infrastructure	13

Frequency Breakdown

Frequency	Count	Percentage
Annually	743	94.3%
Once	24	3.0%
Every 2-3 years	21	2.7%

Priority Breakdown (Top 20 Categories)

Priority	Count
P2 - Important	256
P1 - Critical for food self-sufficiency and dietary diversity	72
P1 - Critical for Infrastructure BS	63
P1 - Critical	44
P2 - Personal Items BS + Score	24
P2 - Liability for balance sheet	23
P1 - Critical for Business BS + Assets + Income	21
P1 - Critical for balance sheet	20
P3 - Contextual	19
P1 - Critical for Land BS + Asset Ownership	19
P4 - Setup only	19
P1 - Critical for Energy BS	16
P1 - Critical for Water BS	13
P1 - Critical for Housing BS + Assets	12
P2 - Business BS (unmet needs)	12
P1 - Critical for Land BS	10
P2 - Storage impacts food security	10
P1 - Critical for Intellectual Wellbeing	9
P1 - Critical for Forestry BS	9
P1 - Critical for Environmental WB + Water BS	8

Key Insights

Strong Coverage Areas OK

- Food System (183 questions): Complete production and consumption tracking feeds Food Balance Sheet and dietary diversity scoring
- Land & Housing (47 questions): Land inventory, housing quality, and asset ownership comprehensively covered
- Resource Domains (118 questions): Water, Energy, Forestry, Mobility, Personal Items, and Business well-documented
- Demographics (63 questions): Family size, income, education, skills captured with some well-being indicators

Gap Areas Warning

Approximately 30 wellbeing survey questions are missing from the current app. These capture subjective individual perceptions:

- Physical: 4 indicators missing: Sleep quality, exercise frequency, energy levels, dental hygiene
- Occupational: 4 indicators missing: Income adequacy, occupational security, emergency fund, debt burden
- Emotional: 2 indicators missing: Anger management, life satisfaction
- Social: 4 indicators missing: Neighbor relations, community contribution, belonging, conflicts
- Environmental: 6 indicators missing: Air quality, green space access, cleanliness, garbage management
- Intellectual: 3 indicators missing: Value of education, learning engagement, information access
- Spiritual: 7 indicators COMPLETELY ABSENT: All spiritual wellbeing questions missing

Sample Annotations by Category

Here are examples showing how different question types were annotated:

Q_ID Range	Category	Relevance	Frequency	Priority
Q0001-Q0019	Registration	ADMIN	Once	P4 - Setup only
Q0020-Q0130	Food Production	Balance Sheet - Food/Income	Annually	P1 - Critical for balance sheet
Q0131-Q0202	Food Consumption	Balance Sheet + Wellbeing(derive)	Annually	P1 - Critical for food self-sufficiency
Q0206-Q0207	Height/Weight (BMI)	Wellbeing - Physical (BMI)	Annually	P1 - Critical for Physical Wellbeing
Q0212-Q0214	Income/Savings	Wellbeing - Occupational + Balance Sheet	Annually	P1 - Critical for Occupational Wellbeing
Q0218-Q0222	Emotional/Social	Wellbeing - Emotional/Social	Annually	P1 - Critical for wellbeing
Q0313-Q0320	Water Quality	Wellbeing - Environmental + Balance Sheet	Annually	P1 - Critical for Environmental Wellbeing
Q0338-Q0350	Energy & Fuel	Balance Sheet - Energy	Annually	P1 - Critical for Energy Score
Q0399-Q0440	Business	Balance Sheet + Wellbeing(derive)	Annually	P1 - Critical for Business Balance Sheet

Recommendations for Malaysia 2026 Deployment

1. 1. Add Missing Wellbeing Questions

- Create a new "Individual Wellbeing Survey" section in the app with ~30 subjective perception questions (sleep, stress, life satisfaction, spiritual practices, etc.)

1. 2. Add Community Infrastructure Section

- Currently a MAJOR GAP — no questions about village-level facilities (schools, hospitals, markets, waste management)

1. 3. Implement Frequency-Based Collection

- Schedule annual cycles for 743 questions, confirm one-time registration for 24 questions, and plan longer intervals for aspirational data

1. 4. Automate Derivable Indicators

- Dietary diversity, asset ownership, and self-sufficiency metrics can be computed from existing data without new questions

— End of Report —

Generated: February 12, 2026

Total Questions Annotated: 788

Output File: Data_Architecture_Map_yk_ANNOTATED.xlsx